

Database Design Document

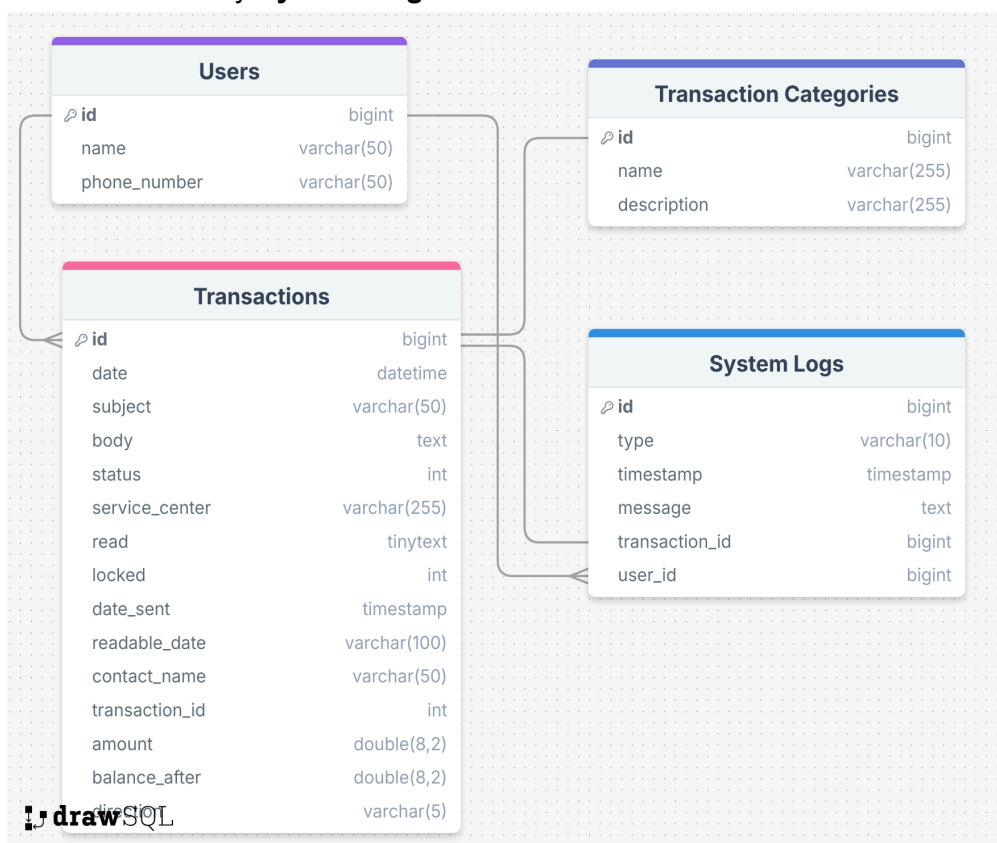
1. Entity Relationship Diagram (ERD)

Entities:

- **Users**
- **Transactions**
- **Transaction Categories**
- **System Logs**

Relationships:

- One **User** → Many **Transactions**
- One **Transaction Category** → Many **Transactions**
- One **Transaction** → Many **System Logs**
- One **User** → Many **System Logs**



2. Design Rationale & Justification

Normalization

- Database follows **3rd Normal Form (3NF)**
- Eliminates redundancy by separating:
 - users
 - transactions
 - categories
 - logs

Scalability

- Logs separated from transactions for high-volume auditing
- Categories table allows dynamic classification without schema changes

Security & Integrity

- Foreign keys enforce referential integrity
- Logs ensure traceability of system actions

3. Data Dictionary

Table: **users**

Column	Type	Description
id	BIGINT PK	Unique user identifier
name	VARCHAR(50)	User full name
phone_number	VARCHAR(50)	User phone number

Table: **transaction_categories**

Column	Type	Description
id	BIGINT PK	Category ID
name	VARCHAR(255)	Category name
description	VARCHAR(255)	Category description

Table: **transactions**

Column	Type	Description
id	BIGINT PK	Transaction ID
user_id	BIGINT FK	Owner of transaction
category_id	BIGINT FK	Transaction category

subject	VARCHAR(50)	Message subject
body	TEXT	Message content
amount	DOUBLE	Transaction amount
balance_after	DOUBLE	Balance after transaction
direction	VARCHAR(5)	IN / OUT
status	INT	Transaction state
locked	INT	Lock status
date_sent	TIMESTAMP	Time sent

Table: `system_logs`

Column	Type	Description
id	BIGINT PK	Log ID
type	VARCHAR(10)	INFO / ERROR
timestamp	TIMESTAMP	Log time
message	TEXT	Log details
transaction_id	BIGINT FK	Related transaction
user_id	BIGINT FK	User responsible

4. Sample Queries

a) Get all transactions for a user

```
SELECT t.*
FROM transactions t
JOIN users u ON t.user_id = u.id
WHERE u.phone_number = '078XXXXXX';
```

b) Transactions by category

```
SELECT t.id, c.name, t.amount
FROM transactions t
JOIN transaction_categories c ON t.category_id = c.id;
```

c) Audit logs for a transaction

```
SELECT *  
FROM system_logs  
WHERE transaction_id = 1001;
```

5. Unique Rules for Security & Accuracy

- Unique Constraints

```
ALTER TABLE users  
ADD UNIQUE (phone_number);
```

- Check Constraints

```
ALTER TABLE transactions  
ADD CONSTRAINT chk_amount CHECK (amount >= 0);
```

- Enum-like Enforcement

```
ALTER TABLE transactions  
ADD CONSTRAINT chk_direction CHECK (direction IN ('IN', 'OUT'));
```

- Indexing for Performance

```
CREATE INDEX idx_transaction_user ON transactions(user_id);  
CREATE INDEX idx_logs_transaction ON system_logs(transaction_id);
```